

HARIDEV NAMBOOD

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SUMMARY

Junior Electrical Engineering student with experience designing circuits and custom PCBs, integrating microcontrollers and sensors, and implementing feedback control systems and computer vision. Seeking an engineering internship for Spring or Summer 2027.

EDUCATION

University of Houston, Cullen College of Engineering | Houston, TX Expected Graduation: May 2028
B.S. in Electrical Engineering **GPA: 3.9**

WORK EXPERIENCE

Robotics Laboratory, University of Houston | Houston, TX May 2026 - Present
Undergraduate Research Assistant

- Developed LabVIEW control software with PLC for miniature medical robots being researched for blood-clot removal in arteries.
- Integrated dual-camera triangulation using OpenCV and path tracking algorithm to achieve >1 mm 3D position-tracking precision.
- Designed and resin-printed 30+ 4 mm microswimmers optimized for controlled magnetic actuation and fluid-dynamics testing.
- Developed a variable-speed motion profile that reduced trajectory-tracking error by approximately 50%, from ~10 mm to ~5 mm.

Army National Guard | Austin, TX Oct 2023 - Present
UH-60M Helicopter Repairer (15T)

- Performed maintenance, inspections, and troubleshooting on 50+ UH-60M Black Hawk helicopters to support aircraft safety and mission readiness.
- Collaborated with a 5-member maintenance team to repair and replace hydraulic pumps, control rods, and rotor blades on UH-60M Black Hawks in support of the 2026 Texas State Mission, following Army technical manuals and maintenance standards.
- Led a squad during phase inspections, coordinating maintenance tasks and documenting aircraft work through the ACN system.

LEADERSHIP

IEEE, University of Houston | Houston, TX Aug 2025 - Aug 2026
President, Micromouse Robotics Club

- Led a 30+ member robotics organization, coordinating technical workshops, meetings, and budget costs throughout the year.
- Mentored members into designing and building their own custom Micromouse robots using C/C++, embedded systems, custom PCB design with KiCad, and 3D enclosures with AutoCAD.
- Mentored 6 teams through robot design, programming, and troubleshooting, achieving a 100% completion rate with all 6 teams building their own functioning robot.

RELEVANT PROJECTS

Micromouse Autonomous Robot | Houston, TX Feb 2025

- Built an ESP32-S3 autonomous maze-solving robot integrating four IR sensors, wheel encoders, IMU feedback, and motor drivers for real-time navigation.
- Implemented a 500 Hz closed-loop motion-control system using encoder and IMU feedback, achieving ≤ 1 mm stopping accuracy and repeatable turning.

Tesla Coil – ASME, University of Houston | Houston, TX Nov 2025

- Co-led a 20-member team in designing and assembling four Tesla coils over two days.
- Built and tested high-voltage circuits while guiding teammates through soldering, circuit assembly, troubleshooting, and high-voltage safety.
- Completed four Tesla coils within two days, with tested circuits producing plasma discharges approximately 10 cm long.

Self-Balancing Robot | Houston, TX Jun 2025

- Built an ESP32-based self-balancing robot with real-time sensor feedback and a tuned PID controller for automatic tilt correction.
- Integrated ESP-NOW wireless control and an LCD for remote operation and real-time system monitoring.
- Maintained balance for 3 minutes with tilt error within $\pm 2^\circ$, recovering from small disturbances within 1 seconds.

TECHNICAL SKILLS

Programming: C/C++, Python, MATLAB, LabVIEW, ARM Assembly

Tools: Microsoft Excel, PowerPoint, Word, KiCad, National Instrumentation Hardware, LTSpice, AutoCAD, 3D Printing, OpenCV

Awards / Honors: High School Valedictorian, Academic Excellence Scholarship, Deans' List (All semesters).